How Much Targeted Investment Does It Take to Top a Multi-Task Leaderboard? Dependence on the Aggregation Rule

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Abstract

Multi-task leaderboards rank models by aggregating their scores on many sub-datasets, and a developer can raise a model's scores on selected sub-datasets by training on related data. We ask how much such targeted investment a given model needs to rank first, and how the amount depends on the aggregation rule. We define investment in score space (sub-datasets touched K, total points L_1, largest single gain L_∞) under explicit caps on every score, compute its minimum exactly under cardinal and ordinal rules, and report the cases where no plan within the caps exists as a feasibility share. On HELM MMLU (79 models, 57 subjects), with scores allowed up to 100, the median model must raise 16 subjects by 696 points in total under the mean; the same model needs 9 subjects and 47 points under plurality, 21 subjects and 112 points under the median, and 47 subjects under Borda. The order of the rules in points is stable across caps (Kendall's $\tau = 0.85$), their order in subjects touched is not ($\tau = 0.63$). If a model can gain at most 10 points per subject, first place is unreachable for 63% of the models under the mean and for every model ranked 31st or lower. On the Open LLM Leaderboard (6 tasks, 4496 models) the minimum is again cheap (paired cost 0.25 of the mean, against 0.15 on HELM), but the saving of the median (0.60, against 0.16) and of plurality (0.50, against 0.07) shrinks and Borda becomes cheaper (0.37, against 0.78), so the order of the rules changes between the two leaderboards (Kendall's τ between them 0.63 in points, 0.46 in tasks touched). The cheapest rules are also the least stable under evaluation noise. Every number comes from the released code, and the assumptions on which the conclusions rest, and where they fail, are stated.

1 Introduction

Multi-task benchmarks and the leaderboards built on them steer which models are adopted, published and funded. A leaderboard gives each model a score on every sub-dataset and then applies an aggregation rule—an average, a median, a count of wins—to produce one ranking. A developer who wants a particular model to be ranked first has more than one lever. One of them is to include data related to some of the sub-datasets in the model's training, which raises the model's scores on those sub-datasets (Zhou et al., 2023; Yang et al., 2023). How much of this *targeted investment* is needed to reach first place, and how that amount depends on the aggregation rule, has not been quantified on real leaderboards.

We study this question on public data without training any model. We (i) define the investment as a change of the target model's own scores on a chosen set of sub-datasets, with a cap on every score, and measure it by the number of sub-datasets touched (K), the total score added (L_1) and the largest single gain (L_∞); (ii) compute, *exactly*, the minimum investment that makes a given model strictly first under each of a set of aggregation rules (cardinal rules from the mean to the minimum; ordinal rules from Borda to the Condorcet winner), for the models of two public leaderboards taken as target (all models, except a stratified sample for the rank rules on the larger leaderboard); and (iii) report where the answer is "impossible" (no plan within the caps works) as a feasibility share rather than hiding it in the averages.

Because what a developer can achieve on a sub-dataset is not observable, the cap on each score is a modelling assumption, and every statement here is made per cap family: from the permissive ceiling of 100 to the current best score on each sub-dataset, to a fixed number of points above the target's own score, to the gains observed between fine-tuned models and their base models on the Open LLM Leaderboard. We state which findings are robust across these assumptions and which are not.

Contributions. (1) A definition of targeted investment in score space with explicit assumptions (single actor, static rivals, no transfer between tasks), and an exact solver for the minimum investment under each rule, validated against closed forms and enumeration (Section 3). (2) A measurement of the investment needed and of its feasibility, per rule and per cap, for all targets of HELM MMLU and for the Open LLM Leaderboard (all targets for the score rules, samples for the rank rules) (Section 4). (3) Robustness checks for normalisation, task granularity, task resampling and evaluation noise, and a statement of when the conclusions fail (Sections 4 and 5). The data, code and one-command reproduction are public (Reproducibility statement).

2 Related work

Fragility of multi-task evaluation. Benchmark outcomes depend on which tasks are included and how they are aggregated: Dehghani et al. (2021) show that the relative standing of methods can change when the benchmark tasks are changed, and Lanctot et al. (2023) view each task as a voter and an aggregate as a social welfare function, comparing rules through their axiomatic properties and empirical robustness. Large-model leaderboards use plain averages: HELM (Liang et al., 2022) reports, among others, accuracy per scenario, and the Open LLM Leaderboard reports the mean of per-task normalised scores, computed with the EleutherAI evaluation harness (Gao et al., 2023). These works vary the *benchmark* (tasks, rule) for fixed models. We instead fix the benchmark and ask what a single *model's developer* must change to win it, and how that cost depends on the rule.

Gaming of leaderboards and contamination. Training on test-related data is a documented route to inflated scores (Zhou et al., 2023); rephrased test samples evade common decontamination filters and let a small model match much larger ones on MMLU and other benchmarks (Yang et al., 2023). Singh et al. (2025) document that undisclosed private testing and unequal sampling distort the Chatbot Arena ranking. These studies show that targeted training on a benchmark's *data* moves scores. Our question is the next one: given that scores on selected sub-datasets can be raised, *how much raising is needed to reach first place*, and under which aggregation rule is that cheapest or impossible.

Strategic post-training. Chen et al. (2026) model a benchmark designer and several developers as a Stackelberg game in which developers allocate resources to benchmark-specific improvements of a latent-quality model, and prove results on equilibria of the induced games. Their object is the equilibrium of strategic developers; ours is the cost of one developer's best response against a static field, measured exactly on observed score matrices. The two are complementary: we do not model rivals' reactions (Section 5).

Social choice. Viewing models as candidates and sub-datasets as voters places the question in computational social choice (Brandt et al., 2016). In that literature, strategic action is usually on the voter side: manipulation of the ballots (Bartholdi et al., 1989) and bribery, i.e. paying voters to change their votes so that a chosen candidate wins (Faliszewski et al., 2009); the cost is the number of voters touched. Here the action is on the candidate side: the target's own task scores change, voters' preferences move as a consequence, and the cost is measured in score points and in tasks touched. Cardinal rules (means, median, minimum) and ordinal rules (Borda, plurality, Condorcet-type) are thus both natural to compare.

3 Setting, investment and aggregation rules

Leaderboard and action. A leaderboard is a score matrix $X \in [0, 100]^{M \times n}$: M models, n sub-datasets ("tasks"), higher is better. One model $i^\star$ is the *target*; all other rows are held fixed (a single actor facing a static field). Targeted investment lets the target raise its score on a chosen task set

$S \subseteq [n]$ to any $y_t \in [x_t, c_t]$ with $x = X_{i^\star,.}$, and leave the other tasks unchanged; $c_t \geq x_t$ is a *cap*. This models "putting data relevant to sub-dataset t into training" as a score gain on t only. It abstracts from the cost of obtaining the gain, from transfer between tasks, and from rivals' reactions; these assumptions and their limits are discussed in Section 5.

First place. An aggregation rule R maps the matrix to a ranking. The target is *first* under R if it is strictly ahead of every other model, with a margin of $\varepsilon = 10^{-3}$ score points on the rule's own scale where the rule is cardinal (ties for first do not count). For rank-based rules ties in a task count half.

Investment: three coordinates. For a plan y we measure the number of tasks touched $K = |\{t : y_t > x_t\}|$, the total score added $L_1 = \sum_t (y_t - x_t)$, and the largest single jump $L_\infty = \max_t (y_t - x_t)$; L_∞ is the cost of a uniform improvement of the target on every task, the natural "broad" baseline against a targeted one. For each (target, rule, cap) we compute the plan that minimises K and, separately, the plan that minimises L_1 (and the smallest uniform raise for L_∞). The minimisers differ in general, so we always say which objective is minimised. If no plan within the caps makes the target first, the cell is *infeasible*; infeasible cells are reported as a share and are never counted as zero or as maximal cost.

Caps (the main modelling assumption). How far a score can be pushed is not observable, so we treat it as a scenario: `ceil` ($c_t = 100$, an upper envelope that is the most permissive), `front0` and `front2` (the best score of any other model on t, plus 0 or 2 points: nothing beyond the current frontier), `gain5`, `gain10`, `gain20` (target's own score plus $5/10/20$ points), and, for the Open LLM Leaderboard only, `emp95` and `emp99`, the 95th and 99th percentile per-task gains of fine-tuned models over their base models in the same leaderboard. Every result is shown per cap; none of the caps is claimed to be the truth.

Aggregation rules. *Cardinal rules* aggregate the score vector of each model: mean (the rule of the Open LLM Leaderboard and the usual headline), geometric and harmonic mean, power means with exponent $p \in \{0.5, -0.5, -2, -5\}$, trimmed mean, median, and minimum (worst task). *Ordinal rules* use only within-task comparisons: Borda count (equivalent to the mean rank), plurality (number of tasks with the highest score), the requirement of being the Condorcet winner (beating every rival in a strict majority of tasks), and Copeland. Being the Condorcet winner is sufficient for first place under every Condorcet-consistent rule (Copeland, Schulze, Kemeny, ranked pairs, minimax) but not necessary for any particular one, so its minimum cost is an *upper bound* on the cost for such rules and is exact only for "first under all of them". We do not compute Kemeny, Schulze or ranked pairs.

Exact solvers. All minima are computed exactly, not by heuristics. For the mean the minimum L_1 has the closed form $n(\tau + \varepsilon) - \sum_t x_t$ when caps do not bind, where τ is the best rival's mean, and the minimum K is a greedy cover of the gap by the largest headrooms $c_t - x_t$. For power means with $p < 1$ the optimum equalises $(y_t + s)$ at a water level; for the minimum and the median, K is the number of tasks that must be lifted above the best rival's value (every task currently below it for the minimum; enough tasks to put a majority above it for the median, at most $\lfloor n/2 \rfloor + 1$). Trimmed mean and the ordinal rules are solved as mixed-integer programmes with lazily generated rival constraints, Borda, the Condorcet winner and plurality are validated against exhaustive enumeration on small subsamples of the real data (Table 2); Copeland and the trimmed mean against brute force and a grid search in the test suite (`code/tests`). A solver failure or time limit is recorded as a failure and never as infeasible (none occurred in the HELM grid).

4 Experiments

4.1 Data and protocol

Leaderboards. We use two public leaderboards, both as frozen snapshots. **HELM MMLU** v1.13.0 (release of 2025-01-10): exact-match accuracy of $M = 79$ models on the $n = 57$ MMLU subjects (Hendrycks et al., 2020; Liang et al., 2022); the subjects have 100 to 1534 test items. **Open LLM Leaderboard** (v2, `contents` dataset, commit `9c09a7c`, last updated 2025-03-20): $n = 6$ tasks (IFEval, BBH, MATH Lvl 5, GPQA, MuSR, MMLU-Pro), where BBH (Suzgun et al., 2022)

and the others are reported as the leaderboard's own normalised scores (rescaled against the random baseline). Of 4576 rows we drop 79 duplicate model names (keeping the best average), leaving 4497 models, and exclude those the leaderboard flags, leaving 4496 models and 4495 targets (the model that is first under the mean is not a target). Checksums, licences and download commands are in the reproduction notes.

Targets, rules, caps. Every model is taken as the target in turn, the others being fixed, so the unit of analysis is the (target, rule, cap) triple. Rules and caps are those of Section 3. Because the exact mixed-integer programmes for the rank rules are expensive on 4496 models, the OLLB rank-rule grids (Borda, plurality, Condorcet winner) use a rank-stratified sample of 57 targets (the 20 best-ranked models and 40 at log-uniformly spaced ranks); the score rules use all 4495 targets. Every rule, including Copeland and the trimmed mean, is run on a top-500 slice (the 500 best models by mean, with a rank-stratified sample of 116 targets; Copeland on 30 targets at cap 100 and on all 116 at base $+10$). Medians over a sample of targets are not comparable with medians over all targets: cross-rule statements on OLLB use the same targets (paired ratios, the top-500 slice).

Summaries and uncertainty. Costs are medians over targets, always with the share of infeasible targets alongside (infeasible targets are never given a cost); confidence intervals are percentile bootstrap intervals over targets (seed 0, 1000 resamples). The task-level and evaluation-noise uncertainty of a single target's cost is studied separately (Section 4.6).

4.2 How much investment, by aggregation rule

Figure 1 and Appendix Table 4 give the minimum investment to first place on HELM MMLU with each of the 78 models that are not already first under the mean rule as target, with the cap at 100 (every task can be raised to the ceiling) and with the cap at the best existing score on each subject. With the cap at 100 every target can reach first place under every rule, so all medians below are over all 78 targets.

The rule changes the price by an order of magnitude. Under the mean the median target must raise 16 of the 57 subjects (95% CI over targets $[13, 18.5]$) by $L_1 = 696$ points in total ($[556, 875]$). Under plurality the median is $L_1 = 47$ points on 9 subjects; under the minimum, 98 points on 11.5; under the median, 112 points on 21; for becoming Condorcet winner, 213 points on 29; under Borda, 534 points on 47 (Appendix Table 4). Paired by target, plurality, the minimum and the median cost 0.07, 0.15 and 0.16 of the mean.

Points and tasks do not rank the rules the same way. Borda costs 0.78 of the mean in points but needs the most subjects (47, 82% of the 57): the Borda credit a target gains on one subject is bounded by the number of models it passes. The median and the minimum need few points but touch 21 and 11.5 subjects, more than plurality (9). The geometric, harmonic and power means lower the points (paired ratios 0.98 to 0.77 from $p = 0.5$ to $p = -5$) but not the number of subjects (15.5–16; Appendix Figure 5). With the cap at the best existing score, the order of the rules in L_1 is almost unchanged (Kendall's τ between caps, median over pairs of HELM caps: 0.85, range 0.63–0.93), the order in K is not (median $\tau = 0.63$, range 0.19–0.87); in K the mean goes from 16 to 37 subjects and plurality from 9 to 18.

Borda and the Condorcet requirement are structural, not accidental. For the median target the Condorcet-winner plan touches 29 subjects at every cap, the majority threshold $\lfloor 57/2 \rfloor + 1$; Copeland, bounded above by it, has the same K and almost the same L_1 (213 versus 213 points at 100; 254 versus 259 at the best score). Five weighted means with Dirichlet$(1, \ldots, 1)$ weights are cheaper than the plain mean: median K 7–9 (mean: 16), paired L_1 ratio 0.36–0.42 (Appendix Figure 4), more than the choice among geometric, harmonic or power means with $p \geq -2$.

One target, five plans. For the model ranked 5 under the mean (Claude 3 Opus), with each subject capped at $+10$ points, an L_1-optimal plan costs 154 points on 16 subjects under the mean, 6.2 points on 5 subjects under the median, 5.0 points on 10 under plurality, 43.7 on 46 under Borda and 24.5 on 27 for the Condorcet winner (Appendix Figure 16; optimal plans are not unique, and the L_1-optimal plan of the ordinal rules touches more subjects than their K-minimal one). The plans do

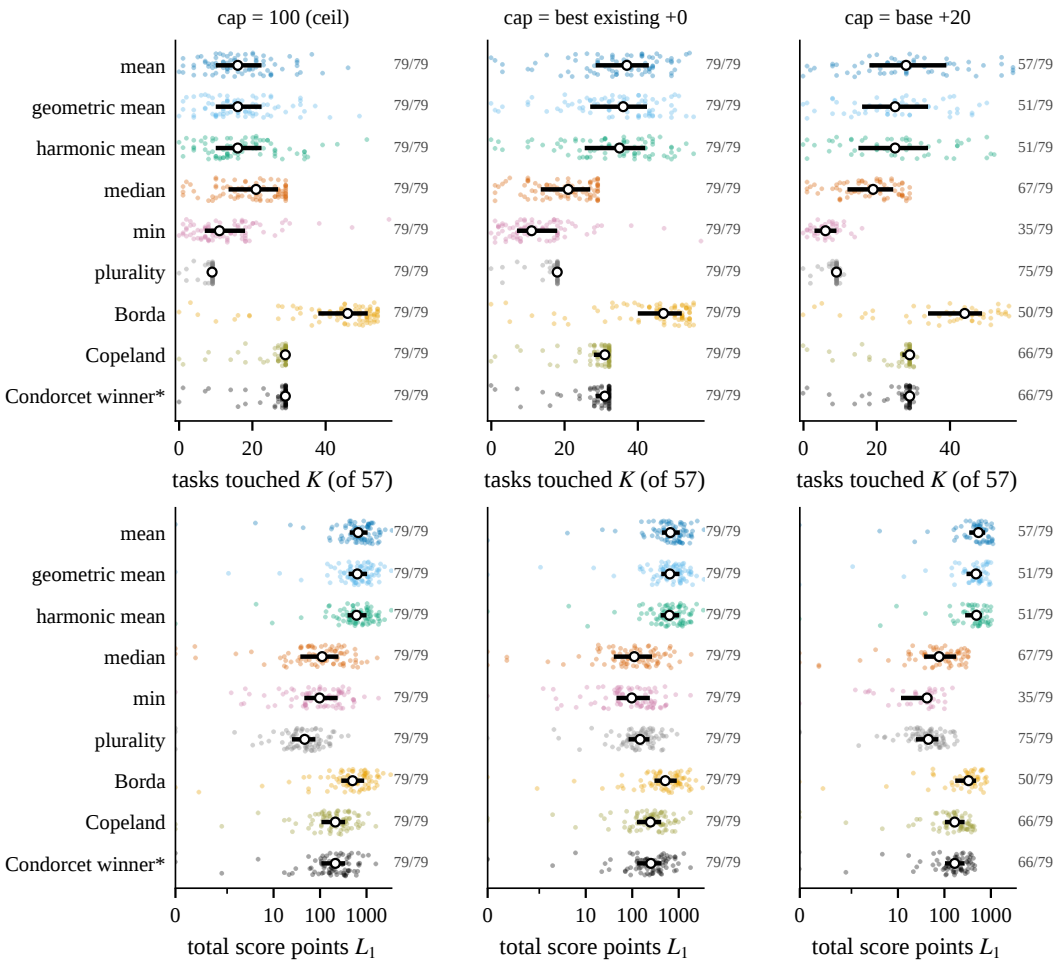


Figure 1: Minimum targeted investment to first place by aggregation rule and cap on each task's score (HELM MMLU, 78 targets, 57 subjects). Top: subjects touched K (minimised); bottom: total points L_1 (minimised separately). Dots are targets; bar: interquartile range; circle: median, over the targets for which first place is reachable (feasible/79 printed at the right; infeasible targets are not drawn and not counted as zero). *Condorcet winner: an upper bound for Condorcet-consistent rules. Caps: 100; best existing score on each subject; target's score $+20$.

not transfer: the mean plan makes the model first under the median and plurality but fourth under Borda; the median and plurality plans leave it fifth under the mean.

4.3 Cost against the target's current standing

The cost grows with the distance to first place (Figure 2). Across targets, L_1 is almost a monotone function of the target's rank under the rule itself (Spearman $\rho = 0.99$ for the median, 0.88 for the minimum, 0.61 for plurality, cap 100) and of its score shortfall to the rule's leader (median $\rho = 0.998$ over 14 rules; lowest 0.88). The slope of $\log L_1$ on $\log(\text{rank})$ at cap 100 is 1.13 ($[0.92, 1.34]$) for the mean, 1.91 ($[1.71, 2.07]$) for the median, 1.59 for the minimum and 3.29 ($[2.21, 4.11]$) for plurality: the rules that are cheapest near the top become expensive faster with rank (the ranks span only 2–79, so we report these as descriptive slopes). Targets close to the top are cheap: for the mean, the rank-2 model needs 4.0 points on 1 subject, ranks 3–5 a median of 42 points on 1 subject, ranks 6–10 177 points on 5 subjects, ranks 11–30 435 points on 10. The headline medians of Appendix Table 4 are dominated by the 49 targets ranked 31 or below, which no developer would pick; we therefore always show cost against rank.

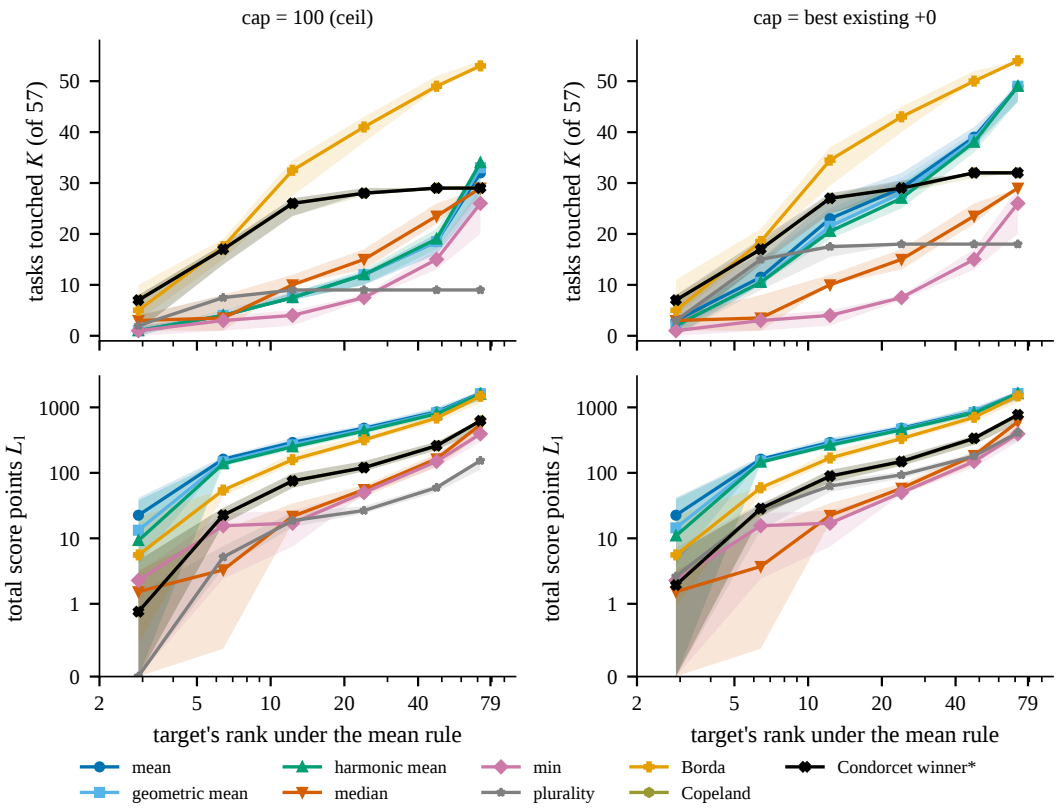


Figure 2: Investment against the target's rank under the mean (78 targets), cap 100 (left) and best existing score (right). Per rank bin: median K (top) and L_1 (bottom, symmetric log) with 95% bootstrap interval of the median (1000 resamples, seed 0). Bins hold 3–32 targets; the interval of the lowest bins is wide. *Condorcet winner.

4.4 Caps and feasibility: when no plan exists

Caps decide whether the answer is a cost or "impossible". With the three permissive caps (100; best existing score $+2$; $+0$) every target is feasible under every rule. With a cap tied to the target's own score the picture changes (Figure 3): the share of targets for which first place is unreachable at $+10$ points is 0.63 (mean), 0.49 (median), 0.27 (plurality), 0.51 (Copeland), 0.68 (Borda) and 0.82 (minimum), at $+5$ points 0.86, 0.79, 0.63, 0.79, 0.86 and 0.91, and at $+20$ points 0.28, 0.15, 0.05, 0.17, 0.37 and 0.56. With a $+10$ gain under the mean, no model ranked 31 or lower can reach first place, whereas 20 of the models ranked 11–30 can, with a median of 44.5 subjects. The minimum is the rule that is cheapest when feasible and the one most often infeasible (82% at $+10$): it demands the target's *worst* subject to exceed the best rival's worst.

Selection effect. Medians under a tight cap are over different feasible subsets (e.g. at $+10$, 29 targets for the mean and 57 for plurality): such a median is not the cost of the same target set, and a rule with fewer feasible targets looks cheaper because only its easy targets are feasible (the minimum at $+10$: 3 subjects, 9 points, for 14 of 78 targets). For cross-rule claims we therefore use the paired ratio on targets feasible under both rules and the ranking at the permissive caps.

4.5 Number of sub-datasets

Does a majority-type rule force about $n/2$ subjects? We drew random subsets of $n \in \{3, 6, 12, 24, 48\}$ HELM subjects (60 draws each) and followed targets at rank 2 and 5 under the mean (Appendix Figure 9). The worst case of the median rule is $K = \lfloor n/2 \rfloor + 1$ (Appendix A) and of the minimum $K = n$, but targets near the top come nowhere near it: at $n = 48$ the rank-5 target

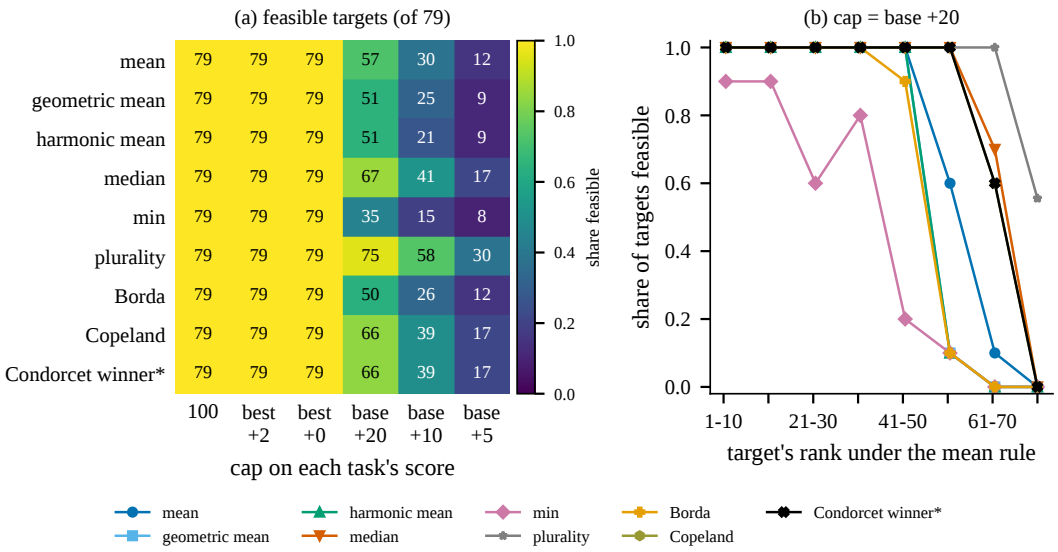


Figure 3: (a) Number of the 79 HELM models for which first place is reachable at all, by rule and cap. (b) Share of targets that can reach first place with the cap at the target's score $+20$, by the target's rank under the mean (bins of ten ranks).

needs $K = 3$ under the median (bound: 25), 4 under plurality and 1 under the minimum. With the cap at $+10$ the mean and Borda grow in proportion to n ($K = 1, 2, 4, 7, 14$ for the mean and $2, 3, 4.5, 7.5, 17$ for Borda at $n = 3, \ldots, 48$, about $0.3n$). The bound is thus tight for a worst-case target and false for contenders: the mean and Borda need a roughly constant share of the subjects, the median, the minimum and plurality a few.

4.6 Robustness to evaluation noise and to the task set

Scores are measured with error on a finite item set. For the ten models ranked 2–11 under the mean, with the cap at $+10$, we redrew every score from its binomial noise (the item count of each subject; 100 draws, seed 0) and, separately, resampled the 57 subjects with replacement (100 draws), and recomputed the minimum investment under seven rules (Appendix Figure 6). The width of the 5–95% interval of L_1 relative to its median is 0.51 (noise) and 0.76 (task resampling) for the mean, but 2.33 and 1.90 for the median, 1.90 and 2.56 for the minimum, 1.49 and 1.51 for plurality, 1.13 and 1.10 for Borda and 1.08 and 1.01 for the Condorcet winner (median over the ten targets, feasible draws). For the minimum, first place is unreachable in 44% of the noise draws and 30% of the task resamples (average over the ten targets). Hence the rules that are cheapest on the observed data are also the least stable: for those rules the 5–95% interval is wider than the median itself. The margin $\varepsilon = 10^{-3}$ is far below the evaluation noise (3.5 points for a subject with the median 173 items at 70% accuracy), so "first" is a statement about the observed table, not about the underlying models.

4.7 Second leaderboard: the Open LLM Leaderboard

The Open LLM Leaderboard has only $n = 6$ tasks and 4495 targets, most of them far from first place (4396 rank below 100 under the mean), so K is small: at cap 100 the median K is at most 3 of 6 tasks for every rule (4 for the Condorcet winner on the top-500 slice); under tight caps the few feasible targets need up to 5 or 6 (Appendix Tables 5 and 6, Figures 10 and 12). Under the mean with the cap at 100, the median target needs 181 points ($[178, 182]$) on 2 tasks; the model ranked 2 needs 4.8 points on one task, the three models ranked 3–5 a median of 7.9 points, those ranked 6–10 27.7 points, all on one task (log–log slope of L_1 on rank 0.44, $[0.43, 0.44]$, against 1.13 on HELM).

Only the minimum transfers. Paired by target at cap 100 (Table 1), the minimum costs 0.15 of the mean on HELM, 0.23 on the top-500 slice (116 targets, every rule on the same targets) and 0.25 on all OLLB targets. The saving of the other rules does not carry over: the median rule costs 0.16,

Table 1: Does the cost order of the rules carry over from HELM to the Open LLM Leaderboard (OLLB)? Cap 100 (every target feasible). Per target set: median paired ratio L_1/L_1^{mean} with 95% bootstrap interval over targets, and median number of tasks touched K. †Run on 57 sampled targets only (ratio to the mean on the same 57). ‡27 targets with a finished solve out of 30 run. *Condorcet winner: upper bound for Condorcet-consistent rules. Last row: Kendall's τ between the rule orders (L_1 / K) of HELM and of the OLLB set, over the rules run in both (including power, weighted and trimmed means not shown).

	HELM (78)		OLLB top-500 (116)		OLLB all targets	
Rule	L_1/L_1^{mean} [95% CI]	K	L_1/L_1^{mean} [95% CI]	K	L_1/L_1^{mean} [95% CI]	K
Mean	1 (ref.)	16	1 (ref.)	1	1 (ref.)	2
Geometric mean	0.96 [0.95, 0.97]	16	0.69 [0.68, 0.70]	1	0.85 [0.85, 0.85]	3
Harmonic mean	0.92 [0.91, 0.94]	16	0.48 [0.46, 0.50]	1	0.70 [0.69, 0.70]	3
Median	0.16 [0.14, 0.19]	21	0.34 [0.30, 0.38]	2	0.60 [0.60, 0.60]	3
Min	0.15 [0.12, 0.18]	11.5	0.23 [0.20, 0.28]	2	0.25 [0.25, 0.25]	3
Plurality	0.068 [0.063, 0.073]	9	0.38 [0.37, 0.41]	2	0.50 [0.45, 0.57]†	2
Borda	0.78 [0.73, 0.82]	47	0.39 [0.37, 0.42]	3	0.37 [0.30, 0.44]†	2
Copeland	0.30 [0.28, 0.31]	29	0.27 [0.15, 0.38]‡	2	not run	
Condorcet winner*	0.30 [0.28, 0.31]	29	0.64 [0.61, 0.65]	4	0.64 [0.58, 0.66]†	3
Kendall τ with HELM, L_1 / K	–		0.46 / 0.49 (18 rules)		0.63 / 0.46 (17 rules)	

0.34 and 0.60, plurality 0.068, 0.38 and 0.50 (the last on the 57 sampled targets), and Borda 0.78, 0.39 and 0.37; on the top-500 slice Copeland costs 0.27 (27 targets with a finished solve) and the trimmed mean 0.82 (Appendix Table 6). Borda, which touches 82% of the subjects on HELM, needs 2–3 of the 6 tasks on OLLB. Kendall's τ between the rule orders of HELM and OLLB is 0.63 in L_1 (17 rules) and 0.46 in K against all OLLB targets, and 0.46 and 0.49 against the top-500 slice; among OLLB settings (caps and variants) the median τ is 0.68 in L_1 and 0.58 in K. The order of the rules is thus a property of the leaderboard as well as of the rules.

Feasibility. With a gain of at most 10 points per task, first place is unreachable for 97% of the OLLB targets under the mean (113 of 4495 feasible), 95% under the median and 90% under the minimum; with the 95th percentile of the observed gains of fine-tuned models over their base models as the cap (`emp95`) it is unreachable for 93% (mean), 97% (median) and 99% (minimum). On the top-500 slice, where the targets are closer to the top, the corresponding shares at $+10$ points are 34% (mean), 31% (median), 25% (minimum), 10% (Borda), 29% (Copeland), 59% (Condorcet winner) and 65% (plurality). The empirical cap is a sensitivity scenario and is not calibrated to HELM (Appendix Figure 13).

Raw against normalised scores. The Leaderboard reports scores normalised against the random baseline of each task. We re-ran the grids on the raw scores and compared matched targets (cap 100). The cost in raw points over the cost in normalised points has median 0.85 for the mean, 0.86 for the median, 0.91 for the geometric mean and 0.99 for the harmonic mean, but 1.38 for the power mean with $p = -5$ and 1.61 for the minimum; the number of tasks touched is the same for 94% (mean) and 29% (minimum) of the targets, and the target's rank under the mean is the same for 1% of the targets (Appendix Figure 14). The two scales are different units, so the cost ratios are not a measure of bias, but their spread (1.9 times between the mean and the minimum) shows that normalisation changes the price of first place by a factor that depends on the rule and moves the order of the rules. Excluding merged models (3788 targets) changes the paired ratios by at most 0.08 (minimum 0.24, median 0.60, geometric mean 0.85, Borda 0.38, Condorcet 0.59, plurality 0.42).

Evaluation noise. For the ten models ranked 2–11 on the top-500 slice (cap base $+10$, 50 draws with binomial noise of each task's item count; seed 0), the width of the 5–95% interval of L_1 relative to its median is 0.52 for the mean, 0.86 for the geometric mean and 0.65 for the Condorcet winner, but 1.83 for the median and 1.59 for Borda (median over targets). First place is unreachable in 52% of the draws under the minimum and in 30% under plurality, and the target's rank under the minimum rule is about 248 in the median draw: that rule is very sensitive to the single weakest cell (Appendix Figure 15).

5 DISCUSSION AND LIMITATIONS

What the evidence supports. On HELM MMLU the aggregation rule changes the price of first place by an order of magnitude, the ranking of the rules by price depends on whether the price is counted in points or in subjects, and the realism of the cap decides whether first place is reachable at all. The cheap rules (plurality, the minimum, the median) are the unstable ones under evaluation noise and task resampling. On the Open LLM Leaderboard the minimum is again cheap, but the savings of the median and plurality shrink, Borda becomes cheaper, the median is again unstable (relative width of the noise interval 1.83), and the order of the rules differs ($\tau = 0.63$ in points, 0.46 in subjects), so we find no single cheapest rule and no order of rules that transfers between leaderboards, nor a law in which the cost scales with the number of sub-datasets (Section 4.5).

Assumptions that carry the results. (i) *Caps.* What a developer can achieve on a sub-dataset is not observable. Cap 100 and the best existing score show the price when scores are free to move, "base $+h$" how quickly first place becomes unreachable when they are not; no cap is claimed to be the truth, and a result is called robust only if it holds across caps. (ii) *Single actor, static field.* Rivals do not react; several developers doing the same would raise the price. (iii) *No transfer, no cost model.* A gain on one subject does not move another, and the cost of obtaining a point is not modelled: L_1 and K are costs in score space, not in data, compute or money. (iv) *Margin.* First place means strictly above every rival by 10^{-3} points, which is far below the evaluation noise of a subject with a few hundred items (Section 4.6); the cost is a statement about the observed table.

Where the conclusions fail or are weak. (1) The order of the rules in K is not stable across caps (τ between 0.19 and 0.87). (2) Medians under tight caps are over different feasible subsets (Section 4.4); only the paired ratios compare rules, and they use only the targets feasible under both. (3) Headline medians are over all targets and are dominated by models far from first place; the cost for a contender (Section 4.3) is much smaller. (4) The Condorcet-winner cost is an upper bound for other Condorcet-consistent rules; we computed only Copeland among them. (5) The weighted means use five random weightings, not any published one. (6) HELM subjects have 100–1534 items each, so single-subject scores are noisy; we studied noise and task resampling for ten targets at one cap (base $+10$) and do not extrapolate to other caps. (7) Only two leaderboards are studied; on the Open LLM Leaderboard the rank rules are sampled and medians over different target sets are not comparable, and the median target is a model far from first place.

Declared incomplete work. The whole analysis took about 15 hours of job time on two CPUs each (`code/REPRODUCE.md`), more than the 4 CPU-hour guidance of the task, although we scaled it down: on the Open LLM Leaderboard Borda, plurality and the Condorcet winner are computed for a rank-stratified sample of 57 targets, and Copeland (30 targets at cap 100, 116 at base $+10$) and the trimmed mean (cost in points only) for a top-500 slice only; 2 of the 30 and 2 of the 116 Copeland programmes reached the 120 s limit and are reported as failed, not as infeasible. The raw-score variants use four caps. We did not compute Kemeny, Schulze or ranked-pairs rules, train any model, or check the single-actor model against real fine-tuning beyond the empirical per-task gains used as caps.

AI USE STATEMENT

This study was carried out by AI agents (large language models) working as separate roles—context and literature review, study planning, analysis and coding, figure and table production, writing, and internal and external review. The agents designed the study, wrote the code, ran the analyses, interpreted the results and wrote this paper. No human author took part in these steps, and we make no claim that a human reviewed them. Every number, table and figure was produced by the code released with the paper; references were checked by the agents against the sources named.

ETHICS STATEMENT

The study analyses publicly released evaluation scores; no human subjects or personal data are involved. It shows how cheaply a leaderboard can be gamed under some rules, which could inform

strategic misuse as well as better benchmark design. We hold that quantifying the vulnerability is useful to benchmark designers and does not enable an attack beyond what training on test-related data already does. The score data carry no stated licence (HELM) or a per-model hub licence column (Open LLM Leaderboard); we use them for analysis and redistribute only derived tables and checksums; see `code/REPRODUCE.md`.

Reproducibility statement

All results are computed from two public snapshots (HELM MMLU v1.13.0; the Open LLM Leaderboard `contents` dataset at a fixed commit), whose checksums and download commands are in `code/REPRODUCE.md` (Table 3). The delivery contains the solvers, the processed score tables, a command for each result grid, all random seeds (bootstrap, noise and task resampling: seed 0; weights: seeds $1000 + k$), the scripts that build every figure and table from the result files (each output has an `*.inputs.json` with the checksums of its inputs), and the solver tests. The exact solvers are validated against closed forms, linear programmes, an independent optimiser and enumeration (Table 2); the runtimes are in the notes.

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A SOLVER BOUNDS AND VALIDATION

Closed forms used. Let τ be the best rival's value under a score rule and x the target's scores. *Mean:* if the caps do not bind, the minimum L_1 is $n(\tau + \varepsilon) - \sum_t x_t$, whatever the allocation, and the minimum K is the smallest number of tasks whose headrooms $c_t - x_t$ sum to at least this gap (a greedy cover by the largest headrooms). *Median:* let m be the best rival's median. The target's median exceeds m as soon as $\lfloor n/2 \rfloor + 1$ of its scores exceed m (for even n both middle values are then above m), so, if the caps allow scores above m, $K \leq \lfloor n/2 \rfloor + 1$; the bound is attained only if all but a minority of the target's scores are below m. *Minimum:* the target's smallest score must exceed the best rival's smallest score, so every task below that value must be raised; $K \leq n$. *Power means ($p < 1$):* at the optimum the lifted tasks share a common water level $y_t + s$ (up to the caps), computed by bisection. The validation in Table 2 checks these against independent references on real data.

Table 2: Validation of the exact solvers (checks are (case, cap, dataset) combinations, not independent instances; error: maximum absolute difference in points, in tasks for K).

Quantity	Reference	checks	max abs. error	all pass
Mean, L_1	linear programme	240	4.5e-13	yes
Mean, L_1 (caps slack)	closed form n (gap $+ \varepsilon$)	240	0.0e+00	yes
Mean, K	greedy headroom cover	240	0.0e+00	yes
Mean, L_∞ (caps slack)	uniform raise $= L_1/n$	240	9.3e-08	yes
Geometric mean, L_1	SLSQP optimiser	240	7.2e-06	yes
Harmonic mean, L_1	SLSQP optimiser	240	3.9e-07	yes
Min and median, K	closed forms	240	0.0e+00	yes
$L_\infty \leq L_1 \leq nL_\infty$	inequality, 4 score rules	240	0.0e+00	yes
Borda, L_1	enumeration, 6 models $\times$ 4 tasks real subsamples	80	1.4e-14	yes
Borda, K	enumeration, 6 models $\times$ 4 tasks real subsamples	80	0.0e+00	yes
Condorcet, L_1	enumeration, 6 models $\times$ 4 tasks real subsamples	80	1.4e-14	yes
Condorcet, K	enumeration, 6 models $\times$ 4 tasks real subsamples	80	0.0e+00	yes
Plurality, L_1	enumeration, 6 models $\times$ 4 tasks real subsamples	80	7.1e-15	yes
Plurality, K	enumeration, 6 models $\times$ 4 tasks real subsamples	80	0.0e+00	yes

Table 3: Data snapshots and numbers of models used.

Snapshot	models used	tasks
Open LLM Leaderboard `contents`, commit `9c09a7c`	4497 models after dropping 79 duplicate names of 4576	6 tasks
HELM MMLU v1.13.0	79 of 79 models with all subjects	57 subjects

B ADDITIONAL FIGURES

C HEADLINE TABLES

D OPEN LEADERBOARD: ADDITIONAL FIGURES

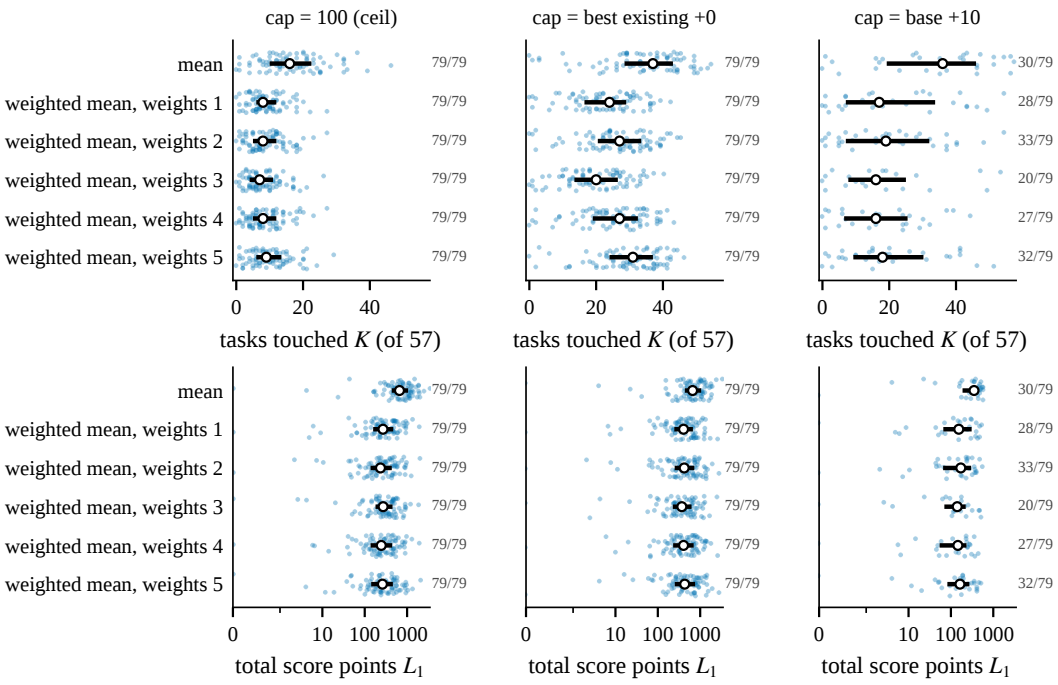


Figure 4: Cost under the unweighted mean and five weighted means (weights from Dirichlet$(1, \ldots, 1)$ over the 57 subjects, seeds $1000 + k$), caps 100, best existing score, base $+10$. Encoding as in Figure 1.

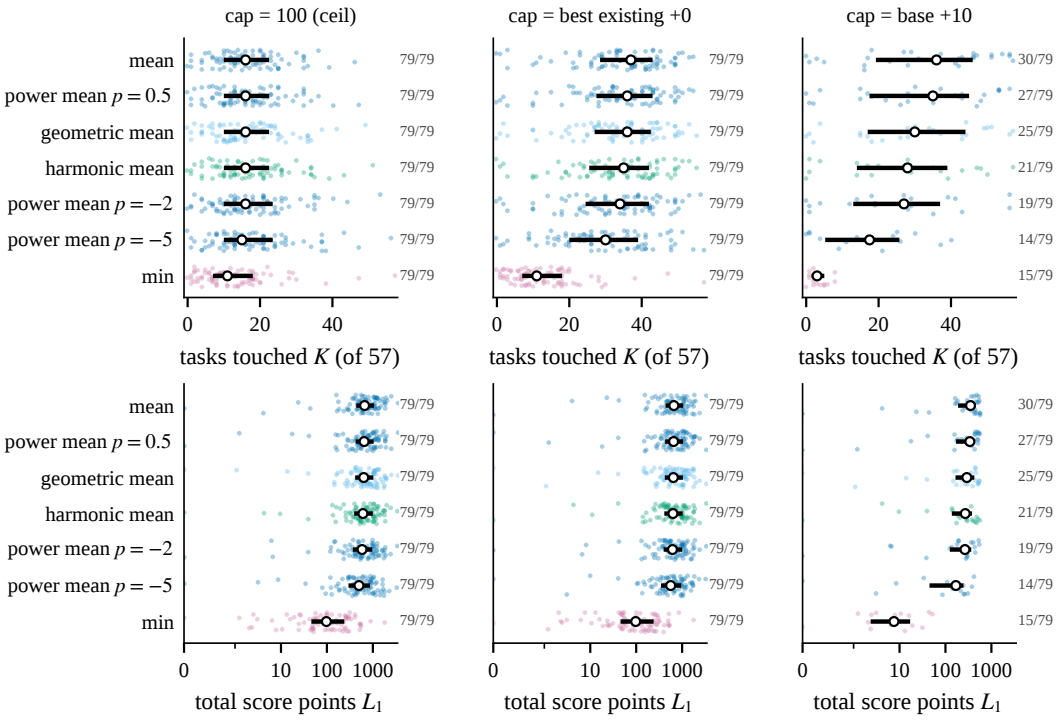


Figure 5: Cost under power means with exponent $p = 0.5, -2, -5$ next to the arithmetic ($p = 1$), geometric ($p \to 0$) and harmonic ($p = -1$) means and the minimum ($p \to -\infty$), caps 100, best existing score, base $+10$. Median K falls as p falls (most clearly at the tighter caps), while the feasible share under base $+10$ also falls.

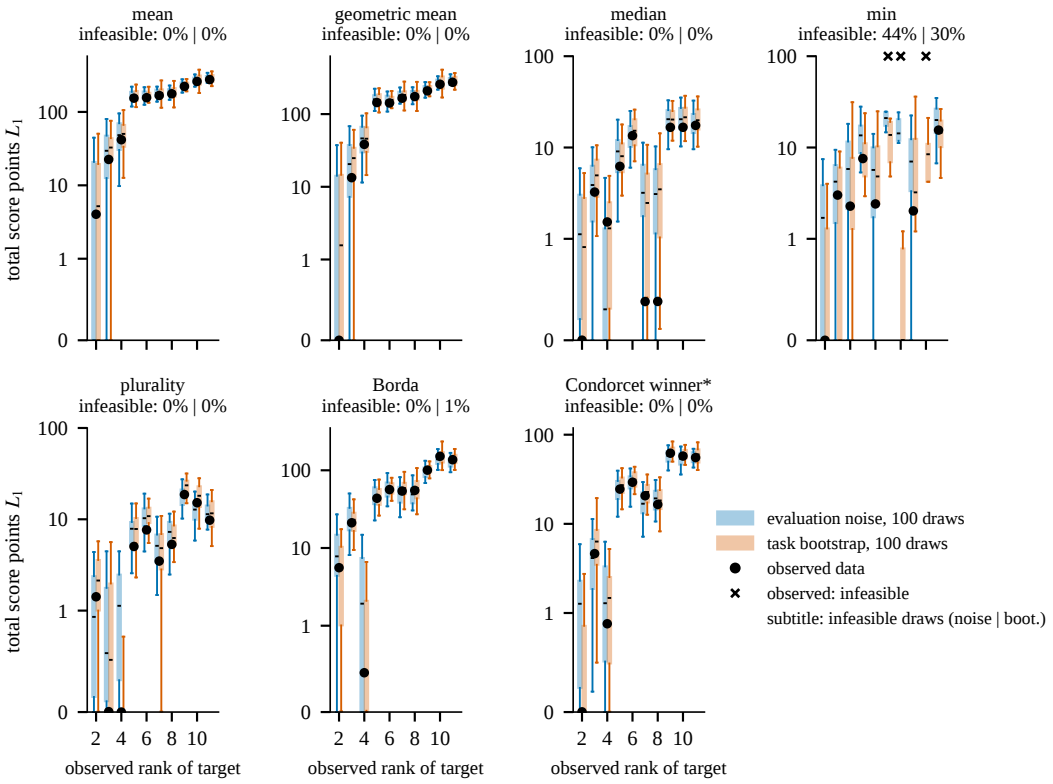


Figure 6: Stability of the minimum L_1 under evaluation noise (blue; 100 draws of every score with binomial noise of its item count, seed 0) and task resampling (orange; 100 bootstrap resamples of the 57 subjects, seed 0) for the ten models ranked 2–11 under the mean, cap base $+10$. Box: interquartile range; line: median; whiskers: 5th–95th percentile over feasible draws; black dot: cost on the observed data, cross: infeasible on the observed data. Subtitles: infeasible share (noise | resampling).

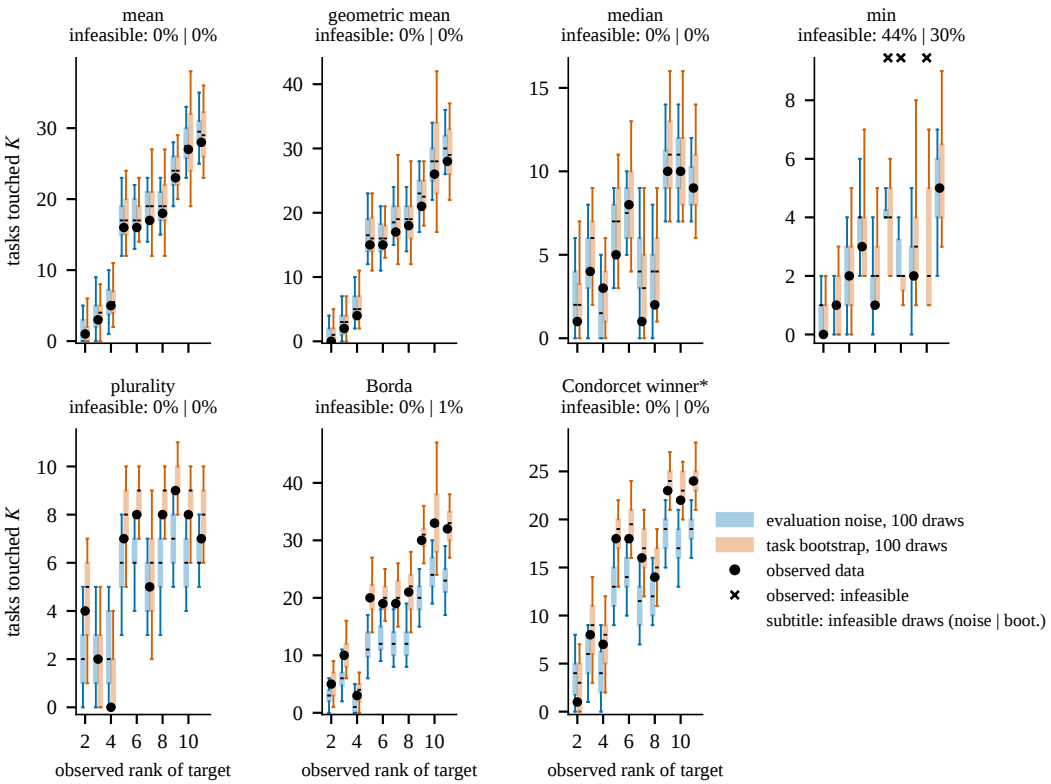


Figure 7: As Figure 6, for the number of subjects touched K.

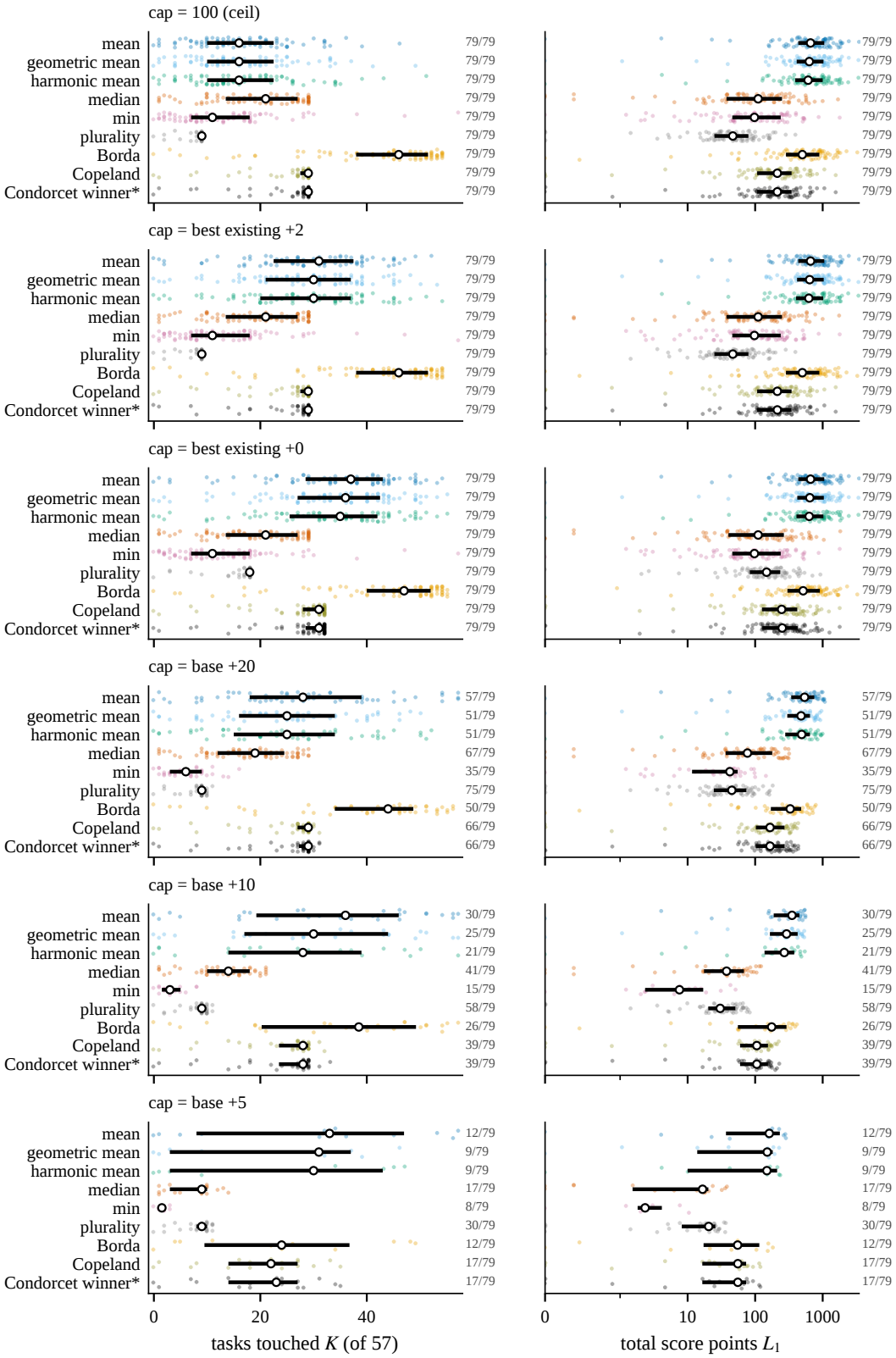


Figure 8: As Figure 1 for all six caps.

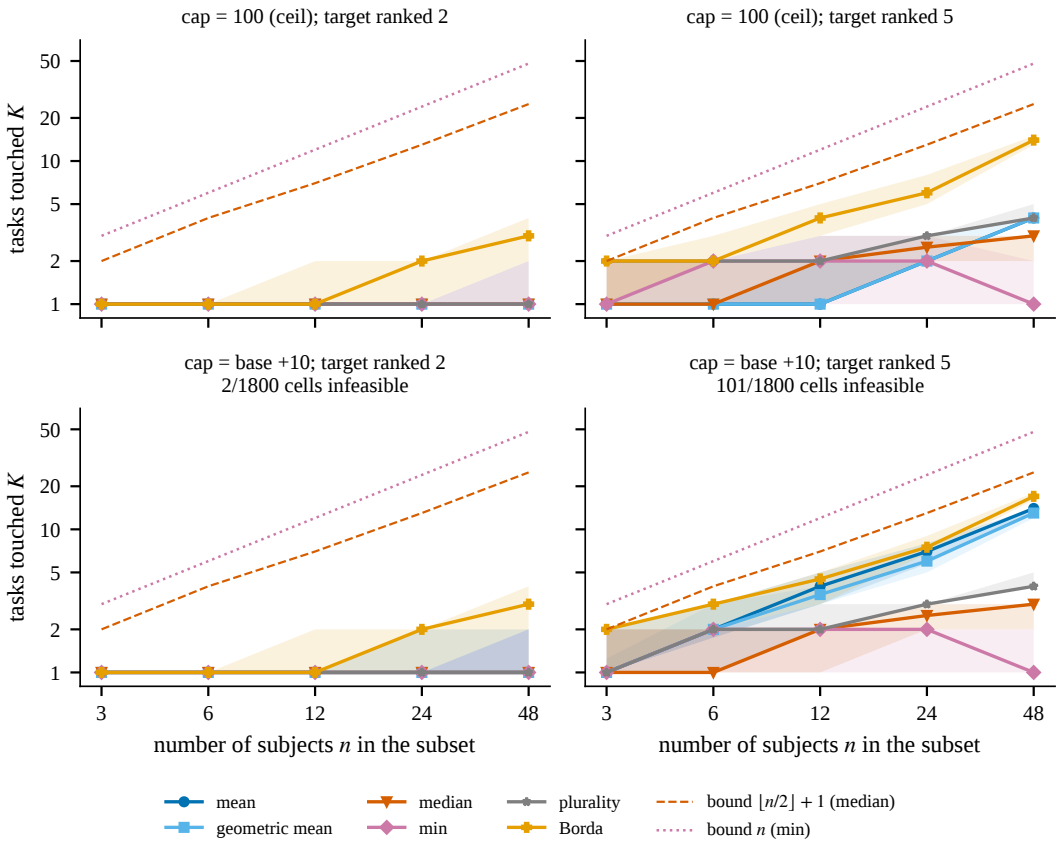


Figure 9: Subjects touched K against the number of subjects n in a random subset of the 57, for targets ranked 2 and 5 under the mean, caps 100 (top) and base $+10$ (bottom). Line: median over 60 draws; band: interquartile range; dashed: $\lfloor n/2 \rfloor + 1$ (median rule); dotted: n (minimum). Infeasible draws are excluded and counted in the panel titles.

Table 4: Headline table (HELM MMLU, 78 targets; feasible-only medians with 95% bootstrap intervals over targets). Under the tight caps (base $+10$, $+20$) each rule's median is over a different set of feasible targets, so comparisons across rows of the K and L_1 columns are conditional on feasibility; the ratio column compares rules on the same targets: median and 95% interval of the per-target ratio of L_1 to the mean rule's, over the n_{both} targets feasible under both ($n_{\text{rule only}}$: targets feasible for the rule but not for the mean, which the ratio cannot use). Infeasible targets are never counted as zero or maximal cost. *Condorcet winner: upper bound for Condorcet-consistent rules.

Rule	feasible	K [95% CI]	L_1 [95% CI]	L_1 / L_1^{mean} [95% CI]	n_{both}	$n_{\text{rule only}}$
cap = 100 (ceil)						
Mean	78/78	16 [13, 18.5]	696 [556, 875]	1 (ref.)	–	–
Geometric mean	78/78	16 [13, 19]	669 [537, 824]	0.96 [0.95, 0.97]	78	0
Harmonic mean	78/78	16 [13, 20]	644 [509, 809]	0.92 [0.91, 0.94]	78	0
Median	78/78	21 [18, 24]	112 [71.7, 163]	0.16 [0.14, 0.19]	78	0
Min	78/78	11.5 [8.5, 15]	98.1 [55.8, 155]	0.15 [0.12, 0.18]	78	0
Plurality	78/78	9 [9, 9]	47.2 [35.1, 59.6]	0.068 [0.063, 0.073]	78	0
Borda	78/78	47 [44, 49]	534 [400, 687]	0.78 [0.73, 0.82]	78	0
Copeland	78/78	29 [29, 29]	213 [162, 259]	0.30 [0.28, 0.31]	78	0
Condorcet winner*	78/78	29 [29, 29]	213 [159, 259]	0.30 [0.28, 0.31]	78	0
cap = best existing +0						
Mean	78/78	37 [33, 40.5]	696 [556, 863]	1 (ref.)	–	–
Geometric mean	78/78	36 [32, 38.5]	680 [544, 848]	0.98 [0.97, 0.98]	78	0
Harmonic mean	78/78	35.5 [31, 38]	667 [531, 833]	0.96 [0.95, 0.97]	78	0
Median	78/78	21 [18, 24]	116 [76.3, 187]	0.18 [0.14, 0.20]	78	0
Min	78/78	11.5 [8.5, 15]	98.1 [55.8, 154]	0.15 [0.12, 0.18]	78	0
Plurality	78/78	18 [18, 18]	149 [112, 182]	0.21 [0.20, 0.22]	78	0
Borda	78/78	48 [46, 50]	549 [413, 705]	0.80 [0.75, 0.84]	78	0
Copeland	78/78	31 [30.5, 32]	254 [198, 336]	0.35 [0.34, 0.38]	78	0
Condorcet winner*	78/78	31 [30.5, 32]	259 [198, 340]	0.36 [0.34, 0.38]	78	0
cap = base +10						
Mean	29/78	37 [28, 46]	360 [276, 440]	1 (ref.)	–	–
Geometric mean	24/78	32 [21, 43]	315 [209, 421]	0.99 [0.98, 1.03]	24	0
Harmonic mean	20/78	28 [17, 38]	273 [173, 364]	1.00 [0.93, 1.03]	20	0
Median	40/78	14 [10.5, 16]	43 [33.8, 60.8]	0.091 [0.072, 0.10]	29	11
Min	14/78	3 [2, 5]	9 [2.7, 18.5]	0.048 [0.018, 0.055]	12	2
Plurality	57/78	9 [9, 9]	31 [25.7, 40.4]	0.055 [0.047, 0.063]	29	28
Borda	25/78	40 [32, 47]	185 [100, 280]	0.61 [0.51, 0.67]	25	0
Copeland	38/78	28 [27, 28]	107 [89.9, 134]	0.25 [0.21, 0.27]	29	9
Condorcet winner*	38/78	28 [27, 28.5]	107 [90.4, 145]	0.25 [0.21, 0.27]	29	9
cap = base +20						
Mean	56/78	28 [23.5, 32]	541 [453, 607]	1 (ref.)	–	–
Geometric mean	50/78	25.5 [21, 29]	490 [405, 558]	0.97 [0.96, 0.98]	50	0
Harmonic mean	50/78	25.5 [20, 29]	500 [405, 549]	0.95 [0.94, 0.96]	50	0
Median	66/78	19 [16, 21.5]	80.1 [57, 119]	0.13 [0.10, 0.15]	56	10
Min	34/78	6.5 [4.5, 8]	44 [23.4, 49.7]	0.093 [0.072, 0.11]	34	0
Plurality	74/78	9 [9, 9]	45.5 [32.4, 52.9]	0.063 [0.058, 0.067]	56	18
Borda	49/78	44 [40, 46]	331 [266, 397]	0.71 [0.64, 0.74]	49	0
Copeland	65/78	29 [28, 29]	169 [134, 218]	0.28 [0.27, 0.29]	56	9
Condorcet winner*	65/78	29 [28, 29]	169 [134, 218]	0.28 [0.27, 0.29]	56	9

Table 5: Headline table for the Open LLM Leaderboard, all targets: 4495 for the score rules, the 57 sampled targets for plurality, Borda and the Condorcet winner (medians over a sample are not comparable with medians over all targets; the ratio to the mean rule is over each rule's own target set). Layout as Table 4. Copeland was not run on this set.

Rule	feasible	K [95% CI]	L_1 [95% CI]	L_1 / L_1^{mean} [95% CI]	n_{both}	$n_{\text{rule only}}$
cap = 100 (ceil)						
Mean	4495/4495	2 [2, 2]	181 [178, 182]	1 (ref.)	–	–
Geometric mean	4495/4495	3 [3, 3]	153 [151, 155]	0.85 [0.85, 0.85]	4495	0
Harmonic mean	4495/4495	3 [3, 3]	125 [124, 126]	0.70 [0.69, 0.70]	4495	0
Median	4495/4495	3 [3, 3]	106 [105, 107]	0.60 [0.60, 0.60]	4495	0
Min	4495/4495	3 [3, 3]	43.2 [42.4, 44.1]	0.25 [0.25, 0.25]	4495	0
Plurality	57/57	2 [2, 2]	22.8 [19.9, 27.5]	0.50 [0.45, 0.57]	57	0
Borda	57/57	2 [2, 3]	12.7 [9.9, 21.8]	0.37 [0.30, 0.44]	57	0
Copeland	not run					
Condorcet winner*	57/57	3 [3, 4]	23.3 [18.2, 32]	0.64 [0.58, 0.66]	57	0
cap = best existing +0						
Mean	4495/4495	4 [4, 4]	181 [178, 182]	1 (ref.)	–	–
Geometric mean	4495/4495	5 [5, 5]	161 [158, 163]	0.89 [0.89, 0.89]	4495	0
Harmonic mean	4495/4495	5 [4, 5]	133 [131, 134]	0.74 [0.74, 0.74]	4495	0
Median	4495/4495	3 [3, 3]	109 [107, 110]	0.61 [0.61, 0.61]	4495	0
Min	4495/4495	3 [3, 3]	43.2 [42.5, 44.1]	0.25 [0.25, 0.25]	4495	0
Plurality	57/57	3 [3, 3]	39.2 [35.2, 45.4]	0.92 [0.82, 1.02]	57	0
Borda	57/57	2 [2, 3]	12.7 [9.9, 21.8]	0.37 [0.30, 0.44]	57	0
Copeland	not run					
Condorcet winner*	57/57	3 [3, 4]	23.3 [18.2, 31.9]	0.64 [0.58, 0.66]	57	0
cap = base +10						
Mean	113/4495	6 [5, 6]	50.9 [47.5, 52.8]	1 (ref.)	–	–
Geometric mean	142/4495	5 [5, 5]	48.4 [43.9, 49.9]	0.84 [0.79, 0.90]	88	54
Harmonic mean	264/4495	5 [4, 5]	39.5 [37.5, 41.2]	0.53 [0.47, 0.58]	78	186
Median	203/4495	3 [3, 3]	18.3 [16.8, 20.4]	0.29 [0.27, 0.31]	86	117
Min	446/4495	2 [2, 2]	10.4 [9.8, 11]	0.22 [0.18, 0.25]	77	369
Plurality	11/57	2 [2, 2]	10.4 [7.2, 16.6]	0.50 [0.31, 0.73]	11	0
Borda	43/57	2 [2, 2]	9.9 [8.6, 12.6]	0.30 [0.25, 0.37]	39	4
Copeland	not run					
Condorcet winner*	29/57	3 [3, 4]	15.4 [11.2, 18.2]	0.55 [0.47, 0.58]	29	0
cap = base +20						
Mean	830/4495	5 [5, 5]	85.8 [83.2, 88.1]	1 (ref.)	–	–
Geometric mean	959/4495	4 [4, 4]	72.4 [68, 75.3]	0.78 [0.77, 0.79]	828	131
Harmonic mean	1157/4495	4 [4, 4]	62 [57.6, 65.3]	0.54 [0.52, 0.56]	826	331
Median	641/4495	3 [3, 3]	30.4 [28.8, 31.9]	0.41 [0.40, 0.42]	603	38
Min	2873/4495	3 [3, 3]	33.7 [33.1, 34.3]	0.21 [0.20, 0.21]	822	2051
Plurality	not run					
Borda	not run					
Copeland	not run					
Condorcet winner*	not run					

Table 6: Headline table for the top-500 slice of the Open LLM Leaderboard: 116 sampled targets, every rule on the same targets. Copeland ran at cap 100 on 30 targets and at base $+10$ on all 116 and was not run at the other caps ("not run"); "(k failed)": solver time-outs, excluded from the feasible counts and the medians. Layout as Table 4.

Rule	feasible	K [95% CI]	L_1 [95% CI]	L_1 / L_1^{mean} [95% CI]	n_{both}	$n_{\text{rule only}}$
cap = 100 (ceil)						
Mean	116/116	1 [1, 1]	52.5 [48.2, 56.4]	1 (ref.)	–	–
Geometric mean	116/116	1 [1, 1]	37.1 [31.8, 39.4]	0.69 [0.68, 0.70]	116	0
Harmonic mean	116/116	1 [1, 1]	26 [23.2, 27.6]	0.48 [0.46, 0.50]	116	0
Median	116/116	2 [2, 2]	18.9 [16.5, 20.9]	0.34 [0.30, 0.38]	116	0
Min	116/116	2 [2, 2]	11.9 [9.8, 14.4]	0.23 [0.20, 0.28]	116	0
Plurality	116/116	2 [2, 2]	20 [18.7, 23.6]	0.38 [0.37, 0.41]	116	0
Borda	116/116	3 [2, 3]	20.2 [16.3, 23.6]	0.39 [0.37, 0.42]	116	0
Copeland	28/30 (2 failed)	2 [2, 2]	9 [4.8, 13.1]	0.27 [0.15, 0.38]	27	0
Condorcet winner*	116/116	4 [3, 4]	31.6 [26.3, 36.9]	0.64 [0.61, 0.65]	116	0
cap = best existing +0						
Mean	116/116	3 [2, 3]	52.5 [48.2, 56.6]	1 (ref.)	–	–
Geometric mean	116/116	2 [2, 3]	37.6 [33, 41.7]	0.73 [0.72, 0.74]	116	0
Harmonic mean	116/116	2 [2, 2]	26 [23.7, 27.9]	0.48 [0.47, 0.51]	116	0
Median	116/116	2 [2, 2]	19.8 [16.5, 21.1]	0.34 [0.30, 0.39]	116	0
Min	116/116	2 [2, 2]	11.9 [9.9, 14.6]	0.23 [0.19, 0.28]	116	0
Plurality	116/116	3 [3, 3]	35.7 [31.1, 39.3]	0.67 [0.65, 0.70]	116	0
Borda	116/116	3 [2, 3]	20.2 [15.5, 23.7]	0.39 [0.37, 0.42]	116	0
Copeland	not run					
Condorcet winner*	116/116	4 [3.5, 4]	31.6 [26.2, 36.8]	0.64 [0.61, 0.65]	116	0
cap = base +10						
Mean	76/116	5 [4, 5]	43.3 [38.1, 47.5]	1 (ref.)	–	–
Geometric mean	78/116	4 [4, 5]	37.3 [32.6, 41.5]	0.83 [0.79, 0.92]	70	8
Harmonic mean	79/116	3 [3, 4]	23.7 [20.4, 31.4]	0.52 [0.44, 0.59]	60	19
Median	80/116	2 [2, 2.5]	14.6 [11.1, 16.6]	0.27 [0.14, 0.30]	62	18
Min	87/116	2 [2, 2]	9.4 [8.4, 10.7]	0.23 [0.19, 0.28]	58	29
Plurality	41/116	2 [2, 2]	12.6 [9.6, 14.7]	0.36 [0.31, 0.39]	41	0
Borda	104/116	3 [3, 4]	19.6 [13.1, 21]	0.32 [0.27, 0.36]	75	29
Copeland	81/116 (2 failed)	3 [3, 4]	16.5 [13.6, 18.3]	0.38 [0.33, 0.39]	71	10
Condorcet winner*	47/116	4 [3, 4]	20.8 [17.2, 23.3]	0.54 [0.47, 0.57]	46	1
cap = base +20						
Mean	116/116	3 [3, 3]	52.5 [47.7, 56.6]	1 (ref.)	–	–
Geometric mean	116/116	2 [2, 2]	37.1 [31.8, 38.8]	0.70 [0.69, 0.72]	116	0
Harmonic mean	116/116	2 [2, 2]	26 [23.3, 27.7]	0.48 [0.46, 0.50]	116	0
Median	114/116	2 [2, 2]	18.3 [16.4, 20.8]	0.34 [0.30, 0.38]	114	0
Min	115/116	2 [2, 2]	11.9 [9.7, 14.5]	0.23 [0.20, 0.27]	115	0
Plurality	116/116	2 [2, 2]	20 [18.4, 23.6]	0.38 [0.37, 0.41]	116	0
Borda	116/116	3 [2, 3]	20.2 [16.3, 23.7]	0.39 [0.36, 0.42]	116	0
Copeland	not run					
Condorcet winner*	114/116	4 [4, 4]	30.7 [25.7, 36.4]	0.64 [0.60, 0.65]	114	0

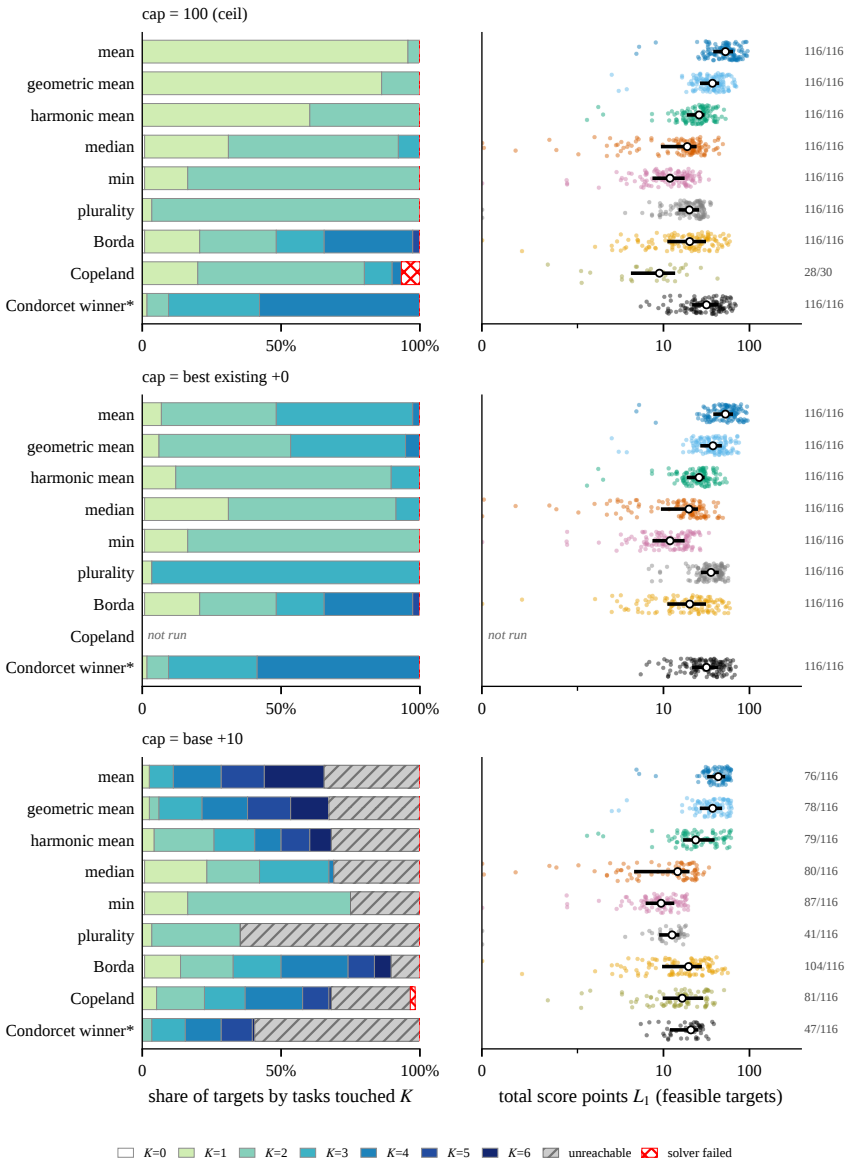


Figure 10: Minimum investment on the top-500 slice of the Open LLM Leaderboard (116 targets, all rules on the same targets), caps 100, best existing $+0$ and base $+10$. Left: share of targets by number of tasks touched K (hatched grey: first place unreachable under the cap; red: solver failed); right: L_1 of each feasible target (symmetric-log axis), bar: interquartile range, circle: median, text: feasible/targets. *Condorcet winner: upper bound for Condorcet-consistent rules.

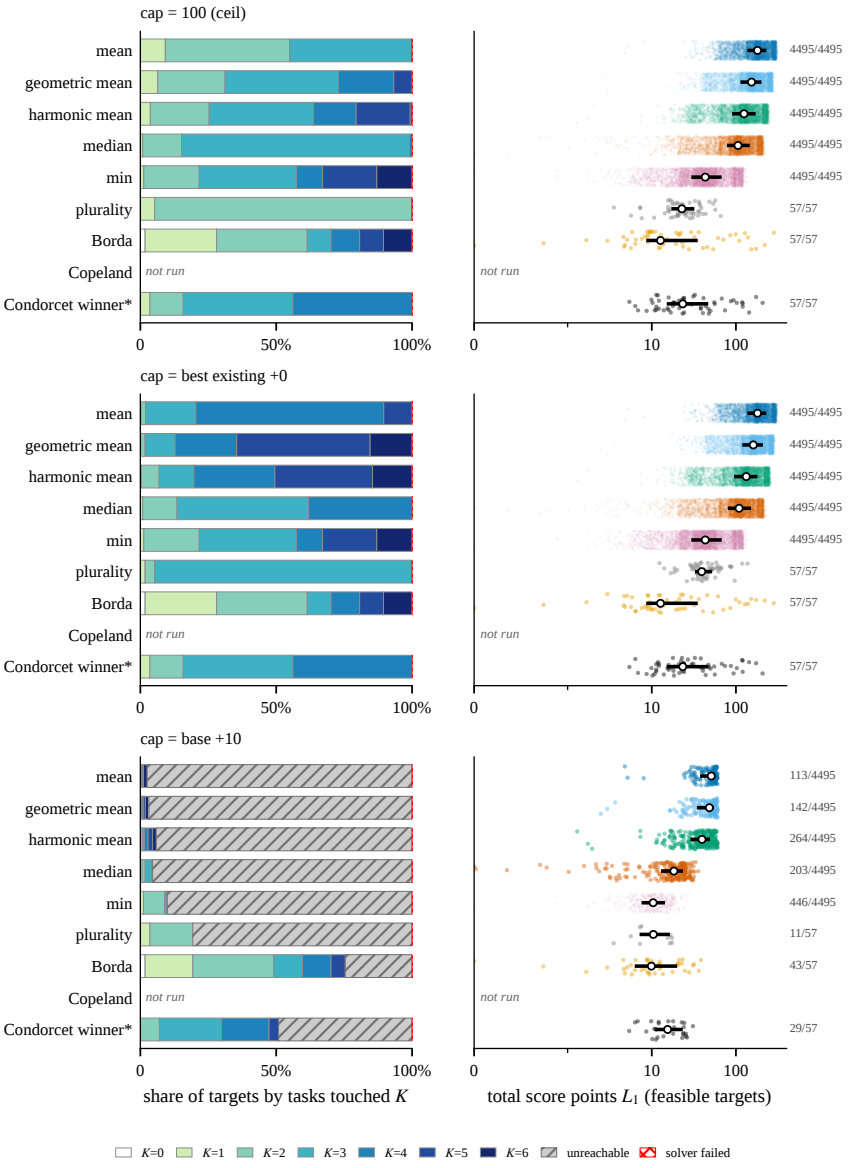


Figure 11: As Figure 10 over all targets: 4495 for the score rules, the 57 sampled targets for plurality, Borda and the Condorcet winner. The two target sets answer different questions and the medians are not comparable across rows.

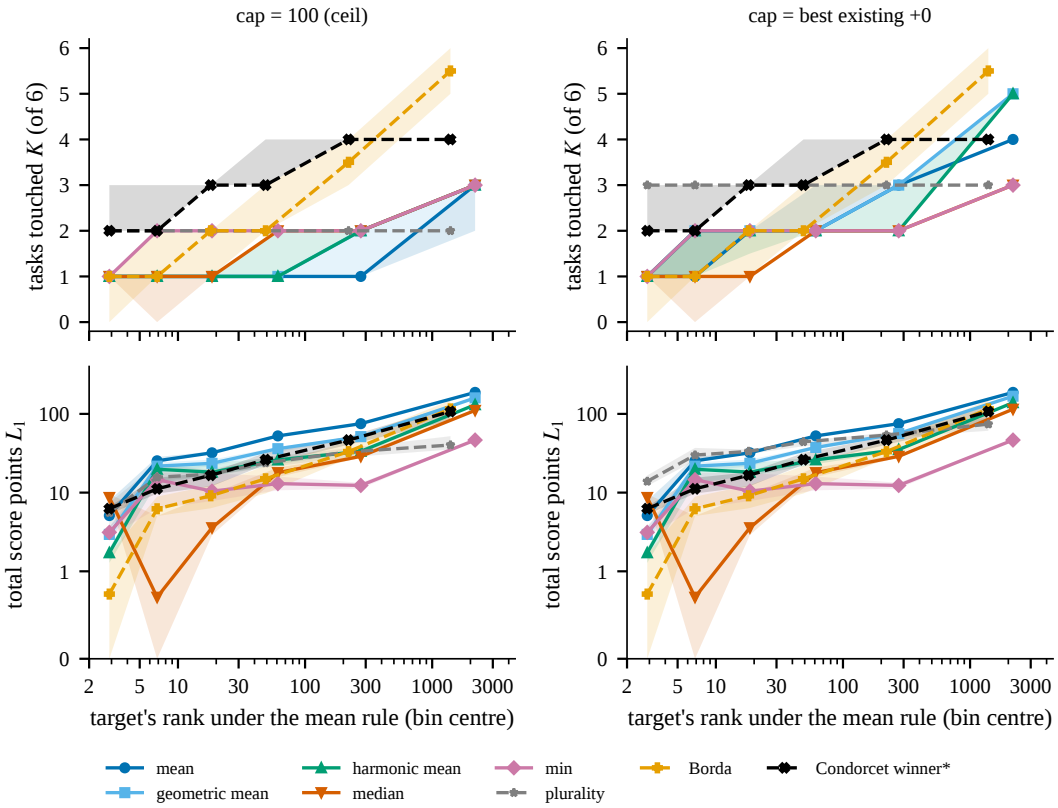


Figure 12: Investment against the target's rank under the mean, Open LLM Leaderboard (4495 targets for the score rules; 57 sampled targets for plurality, Borda and Condorcet, dashed), caps 100 (left) and best existing +0 (right). Rank bins placed at the geometric mean rank; line: median over feasible targets; band: 95% bootstrap interval of the median (1000 resamples, seed 0); bins with fewer than 3 targets are dropped. The rank-rule bins hold 3–19 targets.

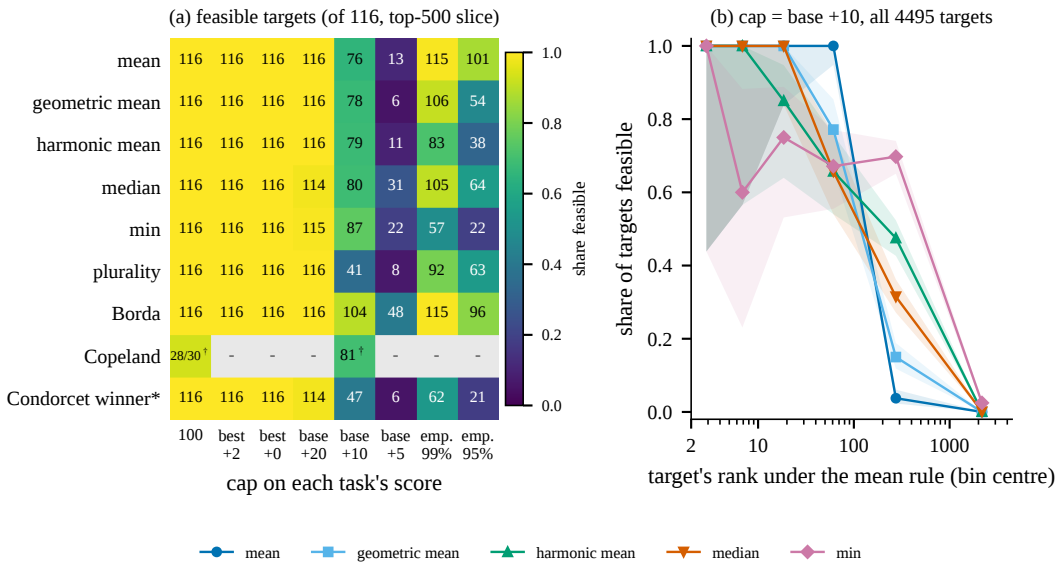


Figure 13: (a) Number of the 116 top-500 targets for which first place is reachable, by rule and cap (−: not run; †: Copeland solver time-outs, counted as neither feasible nor infeasible). (b) Share of the 4495 targets that are feasible under base +10, by rank under the mean, for the five score rules (95% Wilson interval).

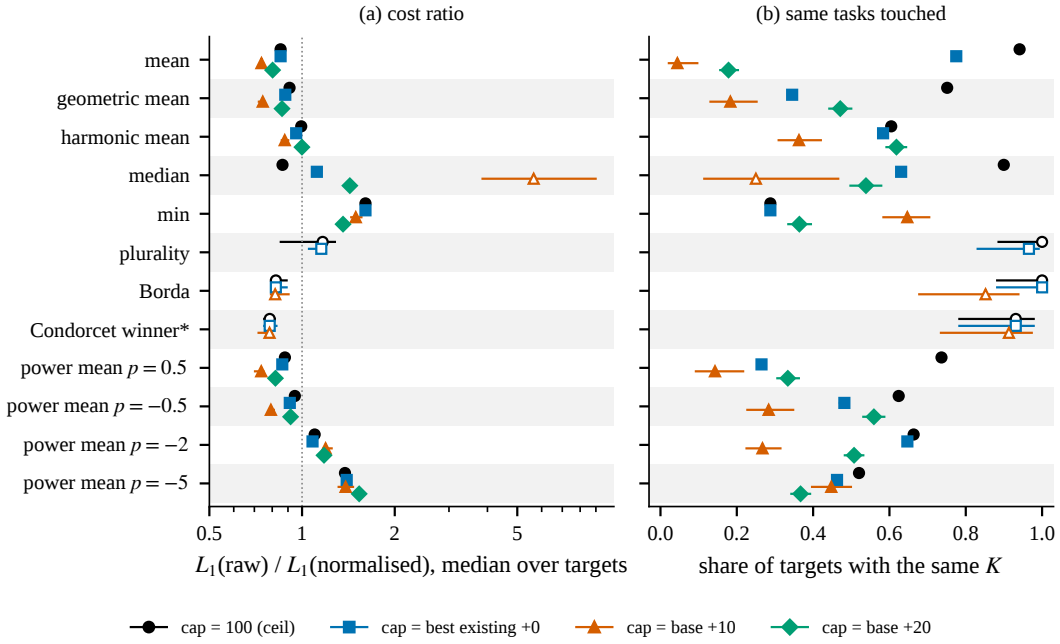


Figure 14: Raw accuracy instead of the Leaderboard's normalised scores. (a) Median over matched targets of L_1(raw)/L_1(normalised), with 95% bootstrap interval (log axis; each cost in points of its own scale). (b) Share of matched targets for which the number of tasks touched is the same in both runs (95% Wilson interval). Matched: feasible and non-zero under both runs; open markers: fewer than 50 matched targets.

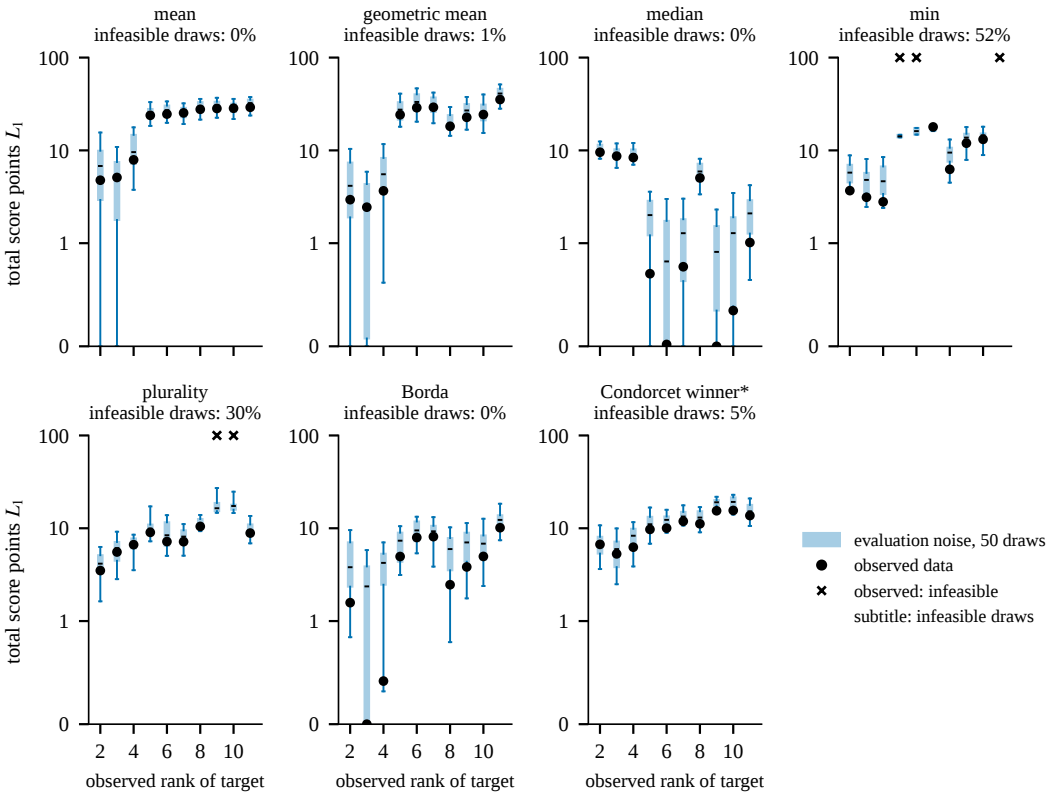


Figure 15: As Figure 6 for the ten top-500 models ranked 2–11 under the mean, cap base +10, evaluation noise only (50 draws, seed 0; the task bootstrap was not run on this leaderboard). Box: interquartile range, line: median, whiskers: 5th–95th percentile over feasible draws; dot: observed cost; cross: infeasible on the observed data.

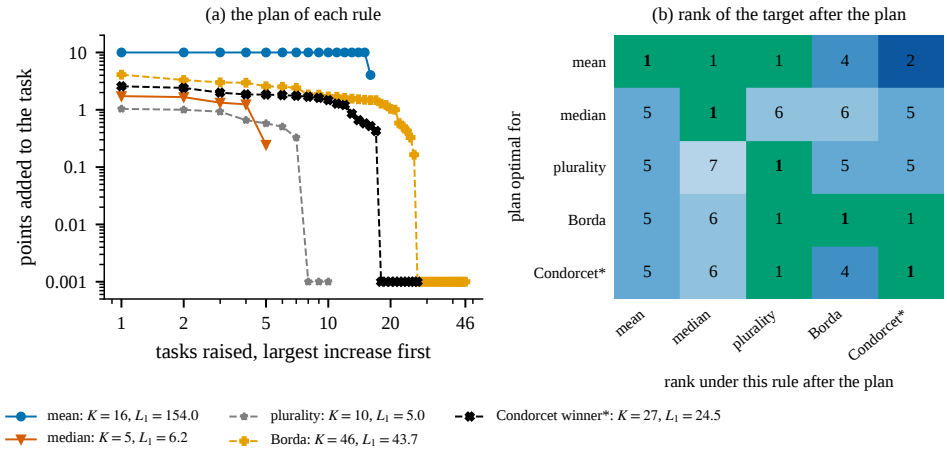


Figure 16: One target, five plans: Claude 3 Opus (20240229) on HELM MMLU, rank 5 under the mean, cap base +10. (a) Points added to each subject by an L_1-optimal plan of each rule (subjects sorted by increase, log axes; legend: subjects touched K and total points L_1; increases of 0.001 are the margin ε). (b) Rank of the target under each rule (columns) after the plan optimal for the rule in the row. One target and one optimal plan per rule (plans are not unique): an illustration, not evidence.